

Module 1: Foundations and Linear Regression

INFO 521 · Machine Learning Foundations · Week 1 · module overview

What predicts systolic blood pressure?

One clinical question opens the whole course. Answering it means choosing a model, writing down what makes a fit good or bad, and solving for the best fit exactly. Every module after this one takes that same machinery and asks harder questions of it.

What the lectures do

Lecture A: Setup, Data, and the Linear Model. NHANES 2021-2022, the leakage rule, the linear model, and mean squared error as the thing you minimise.

Lecture B: Normal Equations and Geometry. The gradient of the loss, the normal equations, least squares as an orthogonal projection, and basis functions when a straight line will not do.

Appendix: Matrix Calculus. The two identities the Lecture B derivation leans on, worked out.

Where this sits in the course

This is the floor. There is nothing behind it to build on, so the module spends its time on things you will use every week after: the notation, the dataset, and the habit of writing down a loss before you fit anything. Module 2 immediately asks whether the fit you just computed means anything.

Week 1 at a glance

- Homework Units 0 and 1. Unit 0 is tooling and reproducibility and is required; Unit 1 is linear models and model selection.
- The prerequisite quiz, worth 1% and gating nothing. It is a readiness check.
- The project is assigned this week. It runs the whole term in two parts, with Part 1 due at the Week 5 midpoint.
- Activity loop: the week01 least-squares explorer, embedded live in Lecture A.
- Discussion: introductions.
- Reading: PML Chapter 1 sections 1.1 to 1.3 and section 1.5; MML Chapters 2 and 9.

Watch out for

The leakage rule is not a formality. Six features were chosen for this dataset specifically because they do not encode the outcome. If you add a feature that does, your model will look excellent and mean nothing.

Solve the system, do not form the inverse. The normal equations are written with an inverse because that is how the algebra reads. In code you solve the linear system. Milestone 1.1 is graded on this.